SKILLS FOCUS 1.1

CONVERTING BETWEEN DENARY AND BINARY NUMBERS

The size of a binary number is referred to by the number of bits it has. A 4-bit binary number will have 4 values that are either 1 or 0, for example, 1001. An 8-bit binary number will have 8 values that are also either 1 or 0. Binary values can get very large, some have as many as 64 bits; now that's a very big number! You will need to be able to understand binary numbers that have as many as 16 bits.

First, focus on converting denary numbers to 4-bit binary. To convert denary numbers to 4-bit binary the units that you need to use are 1, 2, 4 and 8. You can indicate with a 1 or a 0 whether that unit is required. For example, if you convert the denary number 13 to a 4-bit binary number you can use the following steps:

Write down the binary units from 8 to 1. An easy way to do this is using a table.

8	4	2	1

Do you notice how the units increase by the power of 2 each time (going from right to left)?

Compare the denary number (13) to the first binary unit (8) to see whether it is equal to or greater than the denary number. 13 is greater than 8, which means that the unit 8 is required and you can write a 1 beneath it. 8 can then be subtracted from the denary value, $13-8=5$.

8	4	2	1
1			

Now compare the denary number (5) to the next binary unit (4) to see whether it is equal to or greater than the denary number. 5 is greater than 4, which means that the unit 4 is required and you can write a 1 beneath it. 4 can then be subtracted from the denary value, $5-4=1$.

8	4	2	1
1	1		

Then compare the denary number (1) to the next binary unit (2) to see whether it is equal to or greater than the denary number. 1 is not greater than 2, which means that the unit 2 is not required and you can write a 0 beneath it. No subtraction is needed this time as you haven't used the unit.

8	4	2	1
1	1	0	

Finally, compare the denary number (1) to the last binary unit (1) to see whether it is equal to or greater than the denary number. 1 is equal to 1, which means that the unit is required and you can write a 1 beneath it. 1 can then be subtracted from the denary value. $1-1=0$.

8	4	2	1
1	1	0	1

CONTINUED

This means that if you convert the denary number 13 to 4-bit binary, you get the binary number 1101.

Questions

1 What is the largest denary number that can be created with a 4-bit binary number?

2 Convert the denary number 6 to a 4-bit binary number.

3 Convert the binary number 9 to a 4-bit binary number.

To convert denary numbers to 8-bit binary the units that you need to use are 1, 2, 4, 8, 16, 32, 64 and 128. You can indicate, again, with a 1 or a 0 whether that unit is required. For example, if you convert the denary number 150 to an 8-bit binary number you can use the same method as before:

Write down the binary units from 128 to 1:

128	64	32	16	8	4	2	1

Compare the denary number (150) to the first binary unit (128) to see whether it is equal to or greater than the denary number. 150 is greater than 128, which means that the unit 128 is required and you can write a 1 beneath it. 128 can then be subtracted from the denary value, 128 – 150 = 22.

128	64	32	16	8	4	2	1
1							

Next, compare the denary number (22) to the next binary unit (64) to see whether it is equal to or greater than the denary number. 22 is not greater than 64, which means that the unit 64 is not required and you can write a 0 beneath it. No subtraction is needed this time as you haven't used the unit.

128	64	32	16	8	4	2	1
1	0						

Then compare the denary number (22) to the next binary unit (32) to see whether it is equal to or greater than the denary number. 22 is not greater than 32, which means that the unit 32 is not required and you can write a 0 beneath it. No subtraction is needed this time as you haven't used the unit.

128	64	32	16	8	4	2	1
1	0	0					

Now, continue this process until you have compared all the binary units. Remember to subtract the unit from the denary number if it is required. You should end with the following result:

128	64	32	16	8	4	2	1
1	0	0	1	0	1	1	0

CONTINUED

This means that if you convert the denary number 150 to 8-bit binary, you get the binary number 10010110.

A quick check that you can do on your calculation is to look at the last unit on the right of your binary number. If the denary number you are converting is an odd number, this number should be 1. If the denary number you are converting is an even number, this number should be 0.

Questions

4 Convert the denary number 58 to an 8-bit binary number.

5 Convert the denary number 123 to an 8-bit binary number.

If you need to convert a binary number into a denary number, you can use simple addition. You just need to add together all the units that are represented by the value 1.

Try converting the binary number 1010. A simple way to do this is to use a table. You can write the units above and the binary number below:

8	4	2	1
1	0	1	0

You just need to add together all the units that have a 1 below. This means the calculation is $8 + 2 = 10$. Therefore, if you are converting the binary number 1010 to denary, you get the denary number 10.

If you need to convert an 8-bit binary number, you need to extend the units so that you have 8 of them. Remember to increase by the power of 2 each time.

Try converting the binary number 10011001. You can write the values in a table again to make it clearer:

128	64	32	16	8	4	2	1
1	0	0	1	1	0	0	1

You just need to add together all the units again that have a 1 below. This means the calculation is $128 + 16 + 8 + 1 = 153$. Therefore, if you are converting the binary number 100011001 to denary, you get the denary number 153.

Questions

6 What is the largest denary number that can be created with an 8-bit binary number?

7 Convert the 8-bit binary number 01100011 to a denary number.

8 You need to be able to convert up to 16-bit binary values. Fill in the missing units in the table and calculate the denary number for the given binary number.

							256	128	64	32	16	8	4	2	1
0	1	1	0	0	0	1	0	1	0	0	0	1	0	0	1

REFLECTION

How did you know how to add in the correct missing units for Question 8? How did you work out each unit?

Hexadecimal

Hexadecimal is also a number system. Humans can use hexadecimal, but computers cannot process it. Like denary numbers, hexadecimal numbers also need to be converted to binary to be processed by a computer.

Hexadecimal is a base-16 system and uses 16 symbols, these are the numbers 0–9 and the letters A–F. This is to allow the denary values 0–15 to be represented in hexadecimal. This is shown in Table 1.1.

Denary	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

Table 1.1: Hexadecimal representations of denary numbers

The reason for using the symbols A–F for the denary numbers 10–15 is so only one symbol is needed for the denary number in hexadecimal.

SKILLS FOCUS 1.2

CONVERTING BETWEEN DENARY NUMBERS AND HEXADECIMAL

To convert a denary number to hexadecimal, it is easier to convert it to binary first. For example:

First convert the denary number 7 to 4-bit binary.

8	4	2	1
0	1	1	1

To convert the binary number to hexadecimal you can use simple addition again. 4 + 2 + 1 = 7. If you look at the symbol table (Table 1.1), you can see that the hexadecimal for 7 is also 7. Therefore, the denary number 7 is 0111 as binary and 7 as hexadecimal.

So, what happens with denary numbers that are greater than 9?

First convert the denary number 12 to 4-bit binary.

8	4	2	1
1	1	0	0

To convert the binary number to hexadecimal you can use simple addition again. 8 + 4 = 12. If you look at the symbol table (Table 1.1), you can see that the hexadecimal for 12 is C. Therefore, the denary number 12 is 1100 as binary and 12 as hexadecimal.

Converting the denary number to binary first isn't necessary for denary numbers that are less than 16, but it makes the process easier when you need to convert larger denary numbers.

So, how do you convert a larger denary number to hexadecimal?

CONTINUED

First, convert the denary number 201 to 8-bit binary. You know you can use 8-bit binary because the denary number is less than 256.

128	64	32	16	8	4	2	1
1	1	0	0	1	0	0	1

Each hexadecimal symbol only uses 4 bits. Therefore, you need to split the 8-bit binary number into two 4-bit binary numbers.

8	4	2	1
1	1	0	0

8	4	2	1
1	0	0	1

You then convert each 4-bit binary number again using simple addition.

8 + 4 = 12

8 + 1 = 9

The hexadecimal for 12 is C and the hexadecimal for 9 is 9. Therefore, the denary number 201 converted to binary is 11001001 and to hexadecimal is C9.

To convert a hexadecimal number to a binary number of a denary number you can just reverse the process.

To convert the hexadecimal number 5E to a binary number you need to convert each symbol to 4-bit binary. Remember E is the number 14.

8	4	2	1
0	1	0	1

5

8	4	2	1
1	1	1	0

E

Therefore, the hexadecimal 5E converted to a binary number is 01011110.

To convert the hexadecimal 5E to denary, you can use the binary number that you have just calculated.

128	64	32	16	8	4	2	1
0	1	0	1	1	1	1	0

You can add all the units together to get the denary value. 64 + 16 + 8 + 4 + 2 =v94.

The hexadecimal 5E converted to a denary number is 94.

Questions

1 Convert the hexadecimal A2 to a binary number.
2 Convert the binary number 100111010001 to hexadecimal.
3 Convert the denary number 350 to hexadecimal.
4 Convert the hexadecimal 3AC to a denary number.